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Impact of Ease of Use, Usefulness, Attitude, and Trust on AI Adoption Intentions in Higher Education

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Abstract—The integration of Artificial Intelligence (AI) in higher education has attracted increasing attention, particularly in computer education. However, students' acceptance of AI is influenced by several factors that require further investigation. This study aims to examine the effect of perceived ease of use, perceived usefulness, attitude, and trust on the utilization of AI in learning activities among computer education students. A quantitative approach was employed using Partial Least Squares–Structural Equation Modeling (PLS-SEM) with data collected from undergraduate students. The findings reveal that ease of use and usefulness significantly influence students' attitudes toward AI, while trust plays a crucial role in shaping both attitudes and actual utilization. Furthermore, attitude is confirmed as a mediating variable that strengthens the relationship between ease of use, usefulness, trust, and the adoption of AI tools in learning. These results provide empirical support for the Technology Acceptance Model (TAM) and extend it by incorporating trust as an additional construct, offering new insights into AI adoption in higher education. The study highlights both theoretical contributions and practical implications for educators, particularly in the Indonesian context, to integrate AI effectively into learning environments.

***Keywords*—**Technology Acceptance Model (TAM), Trust, AI Adoption, Higher Education, PLS-SEM

I. Introduction

The rapid advancement of Artificial Intelligence (AI) technologies has significantly transformed multiple sectors, including education, where intelligent systems are increasingly integrated to enhance teaching and learning processes. Generative AI tools, for example, have gained considerable attention for their ability to process natural language, generate content, and provide interactive responses, thereby supporting academic tasks such as summarization, translation, and instructional content creation (Supianto et al., 2024). In higher education, the adoption of AI tools is driven by their potential to improve efficiency, foster creativity, and support personalized learning pathways (Bhaskar et al., 2024). Despite these promising capabilities, successful integration of AI in educational contexts requires a comprehensive understanding of the factors that shape user acceptance and trust (Nazaretsky et al., 2025).

Recent studies have examined various dimensions of AI adoption in education. For instance, Afif et al. (2024) demonstrated that AI tools significantly enhance student engagement and academic performance when embedded into university learning environments. Similarly, Finanti et al. (2025) reported that students perceived AI systems as beneficial for academic assignments, but cautioned against the risks of overdependence. Trust has also been recognized as a decisive factor in shaping adoption. Shahzad et al. (2024) Emphasized that without trust, students may hesitate to use AI due to concerns about accuracy, reliability, and ethical implications. These findings resonate with the Technology Acceptance Model (TAM), which identifies perceived ease of use and perceived usefulness as primary determinants of technology adoption by Davis, has been widely applied in recent studies (Bhaskar et al., 2024). This is also reinforced by Theophilus Ehidiamen Oamen (2023) who confirmed that ease of use and perceived usefulness remain central determinants of students' technology acceptance in higher education. While TAM remains a dominant framework in technology adoption research, recent work suggests that additional constructs are necessary to capture the complexity of AI use in education. Scholars argue that trust reduces perceived risks and uncertainties, thereby influencing both intention to use and continued engagement (Mariana et al., 2024; Shahzad et al., 2024). However, studies integrating trust into TAM in the context of AI based education remain limited. Bhaskar et al. (2024); Nazaretsky et al.(2025), for example, identified the importance of trust but did not fully explore its interaction with other TAM constructs such as ease of use, usefulness, and attitude. This indicates a significant research gap.

Other research has explored AI adoption from diverse perspectives, providing valuable but fragmented insights. La Ode Lisbar et al. (2024) analyzed AI integration in blended learning models and found improvements in student comprehension, though challenges with academic integrity persisted. Elfirdaus et al. (2024) applied a simplified TAM framework to AI-based learning platforms, revealing that ease of use strongly influenced perceived usefulness but had mixed effects on student attitudes. Supianto et al. (2024) highlighted the combined influence of perceived usefulness, social influence, and trust on AI acceptance. Beyond AI, innovations such as Augmented Reality-based training Prayogo et al. (2024) and the use of TikTok for enhancing learning achievement C. et al. (2024) further illustrate the rapid digitalization of education. Taken together, these studies underscore the need for models that go beyond traditional TAM constructs and systematically integrate trust.

The novelty of this study lies in extending TAM by incorporating trust into the model and testing it specifically among computer education students. Unlike prior research that either focused on general student populations or examined isolated factors, this study provides a more comprehensive analysis of the direct and indirect effects of perceived ease of use, perceived usefulness, attitude, and trust on behavioral intention to adopt AI. Methodologically, it employs Partial Least Squares Structural Equation Modeling (PLS-SEM) to validate the extended model, providing robust insights into the interactions among constructs.

This research makes important contributions to both theory and practice. Theoretically, it extends TAM by empirically validating trust as both a mediating and moderating variable in AI adoption, addressing gaps noted in earlier studies where trust was acknowledged but not systematically tested (Bhaskar et al., 2024; Nazaretsky et al., 2025). Practically, the findings offer actionable implications for educators, policymakers, and AI developers. By recognizing the factors that enhance acceptance, stakeholders can design AI tools that are not only efficient and user-friendly but also foster strong trust, leading to higher adoption rates and sustainable use. In addition to addressing theoretical gaps, this study also holds practical relevance for Indonesian higher education institutions that are beginning to adopt AI-based learning platforms. By explicitly examining the mediating role of trust, the findings are expected to inform both local practices and global discussions on fostering responsible and effective AI integration in education.

In summary, although AI technologies present vast potential to reshape educational practices, their adoption depends on nuanced understandings of user perceptions and trust. Existing literature provides a strong foundation, but comprehensive models that integrate trust into TAM remain scarce. This study addresses that gap by analyzing the interplay of perceived ease of use, perceived usefulness, trust, and attitude in shaping behavioral intentions toward AI adoption among computer education students. The results are expected to contribute to the design, implementation, and acceptance of AI-based educational tools in higher education.

Building upon these insights and identified gaps, this study advances the Technology Acceptance Model in a more integrative way. Unlike most previous studies that merely extend TAM by adding new variables, this research provides a deeper theoretical perspective by explaining *how* trust operates within the cognitive process of technology adoption. Trust is conceptualized not only as an external construct but as a psychological assurance that reduces uncertainty, strengthens the perception of ease and usefulness, and ultimately shapes a more stable behavioral intention. This conceptual extension contributes to TAM literature by bridging the psychological and technological dimensions of user acceptance. In the specific context of higher education, where AI adoption involves both technical competence and ethical awareness, this study highlights trust as a crucial mediating mechanism that transforms perceived ease of use and usefulness into sustained acceptance. Thus, the model offers new insights for understanding AI adoption that go beyond traditional cognitive evaluations toward relational and ethical dimensions of technology acceptance.

Based on the literature review, it is evident that although the Technology Acceptance Model (TAM) has been widely applied in educational technology studies, limited research has explicitly examined the role of trust in the adoption of AI tools for learning. This study therefore extends TAM by integrating trust and empirically testing its impact among computer education students in Indonesia.

Accordingly, the following hypotheses are proposed:

H1: Perceived usefulness positively affects behavioral intention to use AI.

H2: Perceived ease of use positively affects perceived usefulness.

H3: Perceived ease of use positively affects attitude toward using AI.

H4: Attitude positively affects behavioral intention to use AI.

H5: Trust positively affects attitude toward using AI.

H6: Trust positively affects behavioral intention to use AI.

This study contributes to the literature in three ways:

1. Extending TAM by incorporating trust,
2. Providing empirical evidence on trust's mediating role in AI adoption, and
3. Offering insights for educators on fostering positive attitudes toward AI in higher education.

II. Literature Review

A. Artificial Intelligence in Education

Artificial Intelligence (AI) has become an integral part of modern education, offering innovative solutions that enhance teaching and learning processes. Generative AI tools can analyze large datasets, generate content, and provide personalized recommendations to address diverse learning needs (Dwivedi et al., 2021; Supianto et al., 2024). In higher education, AI is commonly used in tutoring systems, language learning, and automated feedback, all of which increase efficiency and improve student performance (Afif et al., 2024; La Ode Lisbar et al., 2024). However, adoption is not determined solely by technical benefits. Bhaskar et al. (2024) emphasize that user trust and perceived risk also play critical roles, a point reinforced by Akhyar (2023) who showed that AI not only supports learning assistance but also facilitates decision-making through applications such as Naive Bayes algorithms for course selection.

In the Indonesian context, the trend is similar. Jusman et al. (2024) found that AI-assisted platforms significantly improved efficiency in completing assignments, while Fitri Wulandari et al. (2024) reported positive effects on educators' writing literacy. Together, these studies confirm that AI has transformative potential, but successful integration depends on user acceptance and confidence in technology.

B. Technology Acceptance Model (TAM)

The Technology Acceptance Model (TAM), developed by Davis remains one of the most widely applied frameworks to study technology adoption. TAM identifies Perceived Usefulness (PU) and Perceived Ease of Use (PEOU) as central determinants of behavioral intention. PU refers to the belief that technology enhances performance, while PEOU relates to the effortlessness of its use (Purwandani & Syamsiah, 2020). TAM has been validated in numerous educational technology studies, including AI adoption (Bhaskar et al., 2024).

For example, Arif Setia Sandi A et al. (2020) confirmed that both PU and PEOU significantly influence user attitudes in internship systems. Similarly, Elfirdaus et al. (2024) demonstrated that PEOU strongly shapes PU in AI-supported applications. These findings consistently support TAM's proposition that ease of use enhances usefulness, which then fosters positive attitudes and adoption.

C. The Role of Trust in AI Adoption

Despite TAM's explanatory power, it does not explicitly include trust, which has emerged as a critical factor in adoption of AI. Trust refers to the belief that a system will function reliably and securely. In educational contexts, trust reduces risks associated with transparency, bias, and ethical concerns (Nazaretsky et al., 2025; Shahzad et al., 2024).

Several empirical studies underscore its importance. Mariana et al. (2024) identified trust as a key determinant of AI acceptance in secondary education, while Shahzad et al. (2024) argued that sustained AI usage in higher education depends heavily on trust. Nazaretsky et al. (2025) further developed an instrument to measure trust in AI-based tools, showing a strong correlation between trust and intention to use. Collectively, these studies indicate that without sufficient trust, even highly useful and easy-to-use tools may fail to gain traction.

D. Extended TAM with Trust for AI Tools

Despite the robustness of the Technology Acceptance Model (TAM) in predicting user behavior, its classical form has been criticized for underrepresenting social and psychological determinants such as trust. To address this limitation, researchers have extended TAM by incorporating trust. Faiz & Kurniawaty (2023) dan Ramadian & Rahman (2025) demonstrated that trust can serve both as a mediator and moderator, strengthening the effects of PU and PEOU on behavioral intention. In Indonesia, however, empirical evidence on this relationship remains limited. Finanti et al. (2025) highlighted AI's benefits for academic tasks but also noted risks of dependency, while Sugiono (2024) concluded that trust is essential for generative AI diffusion. Setiawan & Luthfiyani (2023) similarly emphasized that fostering trust is as important as ensuring usability in Education 4.0.

Although numerous studies report a positive link between trust and technology adoption, their results remain inconsistent. For instance, Shahzad et al. (2024) found that trust enhances AI adoption in higher education, while Bhaskar et al. (2024) observed that trust does not always exert a direct effect on perceived usefulness. Conversely, Mariana et al. (2024) suggested that the influence of trust may depend on contextual factors such as ethical awareness and digital literacy. These contradictions highlight an unresolved theoretical tension whether trust operates as a direct driver of adoption or indirectly through cognitive perceptions such as ease of use and attitude. Addressing this ambiguity is critical for refining TAM in AI-based education contexts.

Accordingly, this study adopts an extended TAM framework that explicitly incorporates trust into its analysis of AI adoption among computer education students. The proposed model comprises five constructs: Trust, PEOU, PU, Attitude Toward Using (ATU), and Behavioral Intention (BI). By focusing on this population, the study fills an important empirical and contextual gap, offering both theoretical refinement and practical recommendations for designing AI tools that are not only functional and user-friendly but also trustworthy and ethically acceptable for academic use.

III. METHODS

A. Research Design

This study employed a quantitative research design with a survey approach to investigate the influence of perceived ease of use, perceived usefulness, attitude toward using, and trust on students' behavioral intention to adopt artificial intelligence (AI) tools in learning activities. The Technology Acceptance Model (TAM) was adopted as the core theoretical framework and extended by integrating the trust variable to address gaps identified in previous AI adoption studies in higher education (Nazaretsky et al., 2025). The inclusion of trust enriches the classical TAM by capturing psychological and ethical dimensions of technology acceptance, which are increasingly relevant in the context of AI-assisted learning.

The quantitative design was selected because it enables objective measurement of latent variables and statistical testing of hypothesized relationships, ensuring replicable and generalizable findings. Data were analyzed using Partial Least Squares Structural Equation Modeling (PLS-SEM), which is appropriate for examining complex causal relationships and theory extension. The conceptual framework for this study, shown in Figure 1, illustrates the interrelationships among perceived ease of use, perceived usefulness, trust, attitude toward using, and behavioral intention within the extended TAM structure.

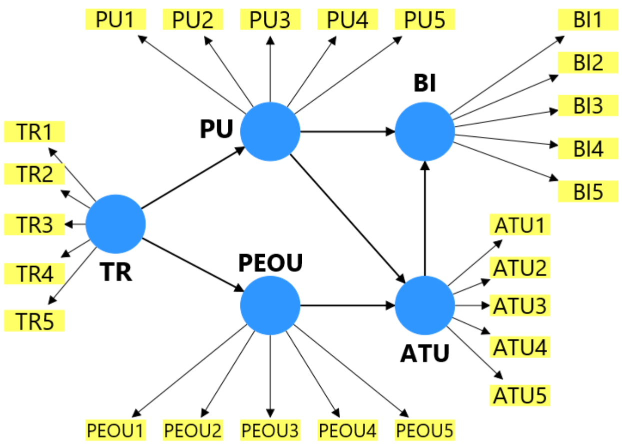

Figure 1. Research Model

B. *Research Stages*

The research process commenced with a comprehensive literature review to establish the theoretical foundation and inform the development of the conceptual model. Subsequently, a structured questionnaire was designed using validated measurement items from previous studies and distributed to respondents via WhatsApp and Google Forms. Data collection was carried out over a period of one week to secure an adequate sample size, after which the responses were meticulously screened to eliminate incomplete or invalid entries, ensuring the reliability and validity of the dataset. The finalized data were then subjected to analysis using Partial Least Squares Structural Equation Modeling (PLS-SEM) to rigorously assess both the measurement model and the structural model.

C. *Variables and Indicators*

The study involved five key variables: Perceived Ease of Use (PEOU), Perceived Usefulness (PU), Attitude Toward Using (ATU), Trust, and Behavioral Intention (BI). Each variable was carefully measured using multiple validated indicators adapted from established TAM and trust-related studies (Purwandani & Syamsiah, 2020; Shahzad et al., 2024). All indicators consistently utilized a five-point Likert scale ranging from "strongly disagree" to "strongly agree," which is widely applied in technology acceptance research. This systematic measurement approach ensures comparability with prior empirical research and provides more reliable and comprehensive insights into students' perceptions of AI adoption.

D. *Population and Sample*

The population of this study consisted of active students of the Computer Education from the 2022 to 2024 cohorts who were enrolled in semesters one to five at the Faculty of Teacher Training and Education, Universitas Mulawarman, totaling 223 students. The cohort distribution was 74 students from the 2022 cohort, 76 students from the 2023 cohort, and 73 students from the 2024 cohort. The sample size was determined using the Slovin formula with a margin of error of five percent, as shown in equation (1):

$$n = \frac{N}{1+N(e)^2} \tag{1}$$

Where n is the sample size, N is the population size, and e is the margin of error. Based on equation (2), the calculation was performed as:

$$n = \frac{223}{1+223(0{,}05)^2} = 143.87 \tag{2}$$

Then it rounded to 144 students. To ensure proportional representation from each cohort, the stratified random sampling technique was applied, and the number of samples from each cohort was calculated using equation (3):

$$ni = \frac{N_i}{N} \cdot n \tag{3}$$

where ni ini is the sample from each cohort, N_i is the population of each cohort, N is the total population, and n is the total sample size. Using this formula, the number of samples from each cohort was determined as 48 students from the 2022 cohort, 50 students from the 2023 cohort, and 48 students from the 2024 cohort. This sampling strategy ensured that all cohorts were adequately represented in the study, allowing for a more accurate and generalizable analysis of AI adoption across different academic years within the program.

E. *Data Collection Techniques*

Data was collected using a structured online questionnaire designed in accordance with the conceptual model. The instrument underwent content validation by experts in educational technology to ensure its relevance and clarity. A pilot test was conducted with a small group of students to refine question wording and format. The use of a Likert scale allowed the measurement of respondents' perceptions in a consistent and quantifiable manner, while the final version of the questionnaire demonstrated satisfactory reliability and validity, ensuring that the data collected accurately reflected the constructs under investigation.

F. *Data Analysis Techniques*

Data analysis was conducted using Partial Least Squares Structural Equation Modeling (PLS-SEM) with the SmartPLS software. The measurement model was evaluated through reliability tests, including composite reliability and Cronbach's alpha, as well as validity assessments using convergent validity (Average Variance Extracted, AVE) and discriminant validity (Fornell–Larcker criterion and HTMT ratio). The structural model assessment further included evaluating path coefficients, t-statistics, p-values, and the coefficient of determination

(R^2) to determine the explanatory power of the model, along with predictive relevance (Q^2) and effect size (f^2) to strengthen the robustness of interpretation. This method was selected because it is particularly suitable for complex models with multiple latent constructs, can handle relatively small sample sizes, and does not require data to meet strict normality assumptions, making it widely adopted in educational technology and social science research.

IV. Results And Discussion

A. Measurement Model Evaluation

The evaluation of the measurement model in this study focused on assessing the reliability and validity of the constructs by using outer loadings, Composite Reliability (CR), Cronbach's Alpha (CA), and Average Variance Extracted (AVE). These criteria are considered essential in determining whether the constructs and their indicators can be trusted to accurately reflect the intended latent variables (Shahzad et al., 2024). According to established thresholds, outer loadings above 0.70, CR and CA values greater than 0.70, and AVE values exceeding 0.50 indicate strong reliability and convergent validity.

As shown in Table 1, all constructions in this study met the recommended criteria, with CA and CR values above 0.70 and AVE values above 0.50. This confirms that the measurement instruments used are both reliable and valid, ensuring that the observed variables appropriately capture their respective constructs (Afif et al., 2024; Alfarobi et al., 2024; Arif Setia Sandi A et al., 2021).

Table 1. Constructing Reliability and Validity

Variable	Cronbach's alpha	Composite reliability		Average variance extracted (AVE)
		(rho_a)	(rho_c)	
ATU	0.933	0.939	0.949	0.789
BI	0.920	0.929	0.940	0.760
PEOU	0.931	0.935	0.947	0.783
PU	0.945	0.947	0.958	0.819

Discriminant validity was examined using both the Heterotrait–Monotrait ratio (HTMT) and the Fornell–Larcker criterion, which help confirm that each construct is distinct from the others. Table 2 shows that all HTMT values are below the 0.90 threshold, satisfying the criterion for discriminant validity (Nazaretsky et al., 2025). These results indicate that the constructions are empirically distinguishable and not affected by multicollinearity issues. Moreover, the Fornell–Larcker criterion further supports this conclusion, as the square root of the average variance extracted (AVE) for each construct exceeds its correlations with other constructs, confirming that each variable captures unique variance. This consistency across multiple indicators provides a strong statistical foundation for evaluating the relationships among the constructs in the subsequent analysis.

Recent methodological research also highlights that the HTMT ratio demonstrates higher sensitivity than the Fornell–Larcker criterion in identifying potential validity problems, particularly in complex structural equation models (Rönkkö & Cho, 2022). Therefore, applying both methods concurrently ensures a more comprehensive and rigorous assessment of the measurement model's quality. Establishing strong discriminant validity is essential for theoretical clarity, as it confirms that constructs such as perceived usefulness, perceived ease of use, trust, attitude, and behavioral intention measure conceptually distinct aspects of AI adoption behavior. Consequently, this result reinforces the reliability and robustness of the extended TAM framework used in this study and provides confidence for proceeding to the structural model evaluation phase.

Table 2. HTMT

Variable	ATU	BI	PEOU	PU	TR
ATU					
BI	0.824				
PEOU	0.031	0.140			
PU	0.874	0.791	0.153		
TR	0.029	0.134	0.851	0.161	

The Fornell-Larcker results in Table 3 confirm the discriminant validity of the measurement model, as the square root of the AVE for each construct ATU (0.888), BI (0.872), PEOU (0.885), PU (0.905), and TR (0.918) is higher than its correlations with any other construct. This indicates that each latent variable shares more variance with its own indicators than with other constructors, demonstrating that the constructions are conceptually distinct. In other words, the measurement items accurately capture the intended theoretical dimensions without overlapping with other variables. These findings provide confidence that the constructions are valid and suitable for examining the hypothesized relationships in the model. Therefore, the model can be considered robust in representing the unique contributions of each construction to the overall theoretical framework.

Table 3. Fornell-Larcker Criterion

Variable	ATU	BI	PEOU	PU	TR
ATU	0.888				
BI	0.771	0.872			
PEOU	-0.014	0.123	0.885		
PU	0.825	0.742	0.142	0.905	
TR	-0.001	0.122	0.809	0.153	0.918

B. *Structural Model Evaluation*

The structural model evaluation aimed to examine the relationships between constructs by analyzing R^2 values, path coefficients, and indirect effects. R^2 values in Table 4 demonstrate the explanatory power of the model. Attitude Toward Using (ATU) achieved an R^2 of 0.699, indicating that Perceived Usefulness (PU) and Perceived Ease of Use (PEOU) collectively explain 69.9% of its variance (Bhaskar et al., 2024). Behavioral Intention (BI) recorded an R^2 of 0.630, meaning that ATU and PU together account for 63% of its variance. Perceived Ease of Use (PEOU) achieved an R^2 of 0.655, explained primarily by Trust (TR). In contrast, Perceived Usefulness (PU) obtained a very low R^2 of 0.023, with TR failing to significantly explain its variance ($p = 0.325$). These values indicate substantial explanatory power for ATU, BI, and PEOU, but limited predictability for PU. This suggests that while the extended TAM effectively explains user attitudes and intentions, additional factors may be required to improve the predictive strength of PU.

Table 4. R-squared Values

Variable	Original sample (O)	T statistics (|O/STDEV|)	P values
ATU	0.699	19.241	0.000
BI	0.630	13.723	0.000
PEOU	0.655	11.340	0.000
PU	0.023	0.988	0.325

Path coefficient analysis, as shown in Table 5, revealed several significant relationships. PU had the strongest positive effect on ATU ($\beta = 0.844$; $p < 0.001$), confirming that students who perceive AI tools as useful tend to develop a favorable attitude toward their use (Jusman et al., 2024). ATU also significantly influenced BI ($\beta = 0.496$; $p < 0.001$), suggesting that a positive attitude increases the intention to use AI in learning activities (Mariana et al., 2024). PU additionally exerted a significant direct effect on BI ($\beta = 0.333$; $p < 0.001$), highlighting its dual role in shaping both attitudes and intentions (Shahzad et al., 2024). Interestingly, PEOU had a significant but negative impact on ATU ($\beta = -0.134$; $p = 0.003$), indicating that excessive simplicity in system operation might reduce students' perception of learning value (Afif et al., 2024; Setiawan & Luthfiyani, 2023). Trust strongly and positively affected PEOU ($\beta = 0.809$; $p < 0.001$), underscoring its role as a key enabler of ease-of-use perceptions (Alfarobi et al., 2024; Supianto et al., 2024). However, TR did not significantly affect PU ($\beta = 0.153$; $p = 0.058$), showing that trust may not directly translate into perceived usefulness (Bhaskar et al., 2024).

Overall, these findings confirm that perceived usefulness remains the dominant predictor of AI adoption intentions, while trust indirectly contributes by enhancing perceptions of usability. The negative influence of PEOU on attitude also introduces an intriguing nuance, suggesting that user-friendly design must still encourage cognitive engagement to sustain positive perceptions in educational contexts.

Table 5. Path Coefficients

Variable	Original sample (O)	P values	Significance
ATU→BI	0.496	0.000	Yes
PEOU→ATU	-0.134	0.003	Yes (Negative)
PU→ATU	0.844	0.000	Yes
PU→BI	0.333	0.000	Yes
TR→PEOU	0.809	0.000	Yes
TR→PU	0.153	0.058	No

The analysis of indirect effects presented in Table 6 reveals several notable mediation pathways, highlighting the complex interplay between perceived usefulness, ease of use, trust, and behavioral intention. The strongest indirect relationship was observed in the pathway PU → ATU → BI ($\beta = 0.419$; $p < 0.001$), confirming that attitude toward using functions as a significant mediator between perceptions of usefulness and students' intention to use AI tools, as supported by Davis and recent findings by (Jusman et al., 2024). Trust also exerted an indirect influence on BI through the pathway PEOU → ATU → BI ($\beta = -0.054$; $p = 0.008$), as well as through a longer mediation chain TR → PU → ATU → BI ($\beta = 0.064$; $p = 0.063$), although the latter did not reach statistical significance, indicating a more nuanced role of trust in shaping behavioral intention (Alfarobi et al., 2024; Supianto et al., 2024). Interestingly, the pathway PEOU → ATU → BI showed a small but significant negative effect ($\beta = -0.067$; $p = 0.006$), which aligns with previous observations that excessive ease of use may reduce motivational engagement or perceived challenge in learning contexts (Setiawan & Luthfiyani, 2023). Furthermore, trust influenced ATU indirectly through PEOU ($\beta = -0.109$; $p = 0.004$), reinforcing the finding that trust can have complex, sometimes counterintuitive effects on attitudes toward technology use.

Taken together, these findings underscore the importance of examining indirect effects to fully capture the mechanisms driving AI adoption. They demonstrate that attitude functions as a central cognitive–affective bridge linking perceived usefulness, ease of use, and behavioral intention, while trust operates as an underlying relational mechanism that enhances or modifies these perceptions. This pattern provides empirical support for extending TAM with trust, as it captures both cognitive evaluations and affective judgments that shape user acceptance in educational contexts. By identifying these indirect pathways, the study contributes to a deeper theoretical understanding of how psychological and technological dimensions jointly influence students' behavioral intentions to adopt AI tools. Overall, these insights highlight that fostering trust and meaningful engagement may be as critical as improving usability or perceived usefulness in ensuring sustainable AI adoption in higher education.

Table 6. Specific Indirect Effect

Variable	Original sample (O)	P values	Significance
TR→PU→ATU	0.129	0.062	No
TR→PEOU→ ATU	-0.109	0.004	Yes (Negative)
TR→PU→BI	0.051	0.115	No
PEOU→ATU→BI	-0.067	0.006	Yes (Negative)
PU→ATU→BI	0.419	0.000	Yes
TR→PU→ATU→ BI	0.064	0.063	No
TR→PEOU→ ATU→BI	-0.054	0.008	Yes

These results align with previous research that highlights the central role of perceived usefulness in technology acceptance models Bhaskar et al. (2024) and the increasingly recognized function of trust in enabling ease of use perceptions (Shahzad et al., 2024; Supianto et al., 2024). They also reinforce findings from Ali et al. (2022); Setiawan & Luthfiyani (2023) regarding the potential negative impact of over-simplicity in educational AI systems.

C. Discussion

The findings of this study provide several important insights into the acceptance of AI-based learning tools within higher education, particularly among computer education students. First, perceived usefulness (PU) emerged as the most influential factor in shaping both attitudes toward using (ATU) and behavioral intention (BI). This result aligns with the central premise of the Technology Acceptance Model (TAM), which posits that when users perceive technology as performance-enhancing, they are more inclined to adopt it. The strong and significant path from PU to BI ($\beta = 0.333$; $p < 0.001$) reinforces the dual role of usefulness in influencing both attitudinal and intentional dimensions, consistent with findings by Shahzad et al. (2024) and other recent TAM-based studies on educational technology.

Interestingly, perceived ease of use (PEOU) exhibited a significant but negative relationship with ATU ($\beta = -0.134$; $p = 0.003$). This unexpected result suggests that when an AI tool is perceived as overly simple, students may question its intellectual depth or capacity to stimulate critical thinking. Similar anomalies have been observed by Afif et al. (2024) and (Setiawan & Luthfiyani, 2023), who argued that in cognitively demanding contexts, simplicity may undermine perceptions of learning value. This finding challenges the conventional assumption of TAM that greater ease automatically increases positive attitudes, revealing that "ease" can, under certain circumstances, reduce engagement and curiosity, an emerging paradox in modern TAM applications.

Trust (TR) was found to exert a strong positive effect on PEOU ($\beta = 0.809$; $p < 0.001$), confirming its role as a psychological enabler of usability perceptions (Alfarobi et al., 2024; Supianto et al., 2024). However, its non-significant direct effect on PU ($\beta = 0.153$; $p = 0.058$) indicates that while trust makes systems feel easier to use, it does not immediately enhance perceptions of usefulness. This aligns with Bhaskar et al. (2024) and (Mariana et al., 2024), who emphasized that trust often functions indirectly, moderating or mediating the effects of other cognitive factors on behavioral intention. Such a nuanced relationship enriches our understanding of the complex psychological pathways that underlie AI adoption decisions.

From a theoretical standpoint, these results extend TAM by empirically validating trust as both a cognitive and affective construct that bridges usability perceptions with behavioral intentions. The discovery of a negative PEOU–ATU linkage contributes a novel insight to the international discussion on AI-based learning, implying that the value of "effort" and "challenge" remains significant in academic motivation. This resonates with global pedagogical frameworks, such as those advanced by UNESCO (2023) and OECD (2024), which advocate for AI designs that not only simplify learning processes but also preserve intellectual rigor and ethical awareness. Theoretically, this underscores that technology acceptance in education is not purely instrumental but deeply intertwined with motivation, autonomy, and moral trust.

The mediation analysis further reveals the multi-layered nature of trust. The indirect path TR → PEOU → ATU → BI ($\beta = -0.054$; $p = 0.008$) suggests that trust shapes ease-of-use perceptions, which subsequently influence attitudes and behavioral intentions. The presence of negative coefficients in these mediating chains implies that high trust may reduce perceived complexity but also risk oversimplification, potentially lowering learners perceived challenge. Hence, system designers and educators must strike a balance between usability and cognitive engagement.

In practical terms, these findings carry significant implications for both local and global educational contexts. In Indonesia, where AI adoption in higher education is still emerging, building and maintaining trust through transparency, reliability, and contextual alignment will be crucial for sustainable implementation. Internationally, this study contributes to the ongoing debate about how ethical, explainable, and participatory AI practices can enhance educational integrity and user confidence. By situating the results within this broader discourse, the study demonstrates that AI adoption is not merely a technical phenomenon but a socio-ethical transformation that requires careful human-centered design.

PU continues to emerge as the strongest determinant of AI adoption, whereas *trust* plays a central psychological role by indirectly influencing users' perceptions of usability and acceptance. The unexpected relationship between PEOU and user attitudes provides a deeper insight into the Technology Acceptance Model (TAM), suggesting that simplicity by itself may not guarantee sustained engagement in higher education contexts. This perspective advances the extended TAM as a more holistic framework for understanding AI adoption, incorporating not only cognitive and affective factors but also ethical considerations relevant to contemporary learning environments.

V. Conclusion

This study demonstrates that perceived usefulness (PU) plays the most central role in shaping both students' attitudes and their intention to adopt AI in higher education. Attitude toward using (ATU) emerges as an important link that connects students' perceptions of usefulness with their behavioral intention. Although perceived ease of use (PEOU) shows a negative direct effect on attitude, it continues to exert influence through indirect pathways that contribute to strengthening students' intention to use AI-based tools.

The inclusion of trust (TR) within the Technology Acceptance Model offers additional clarity regarding how students form judgments about AI systems. Trust does not directly increase perceived usefulness, yet it provides a psychological basis that enhances perceptions of usability and indirectly affects both attitude and intention. These findings indicate that AI adoption is shaped not only by functional considerations but also by users' sense of comfort, confidence, and security when interacting with the system.

From a theoretical perspective, this study enriches the TAM framework by demonstrating the role of trust as a mechanism that supports cognitive evaluations of ease of use and intention. The model broadens TAM beyond its traditional focus on performance-related beliefs by integrating dimensions related to emotional assurance and ethical perception, which are increasingly relevant in discussions about responsible and human-centered AI in education.

The practical implications highlight the importance of strengthening trust to support successful AI integration. Higher education institutions and AI developers can achieve this by improving transparency regarding how AI tools operate, providing continuous training in digital literacy and AI ethics, and conducting routine evaluations to minimize bias and ensure reliability. These efforts can contribute to building an environment where AI is perceived as dependable, fair, and supportive of learning needs.

Further research may extend this model to other academic fields to examine whether the patterns observed in this study apply across disciplinary contexts. Investigating moderate factors such as perceived risk, ethical awareness, or levels of digital literacy may also offer broader insight into the conditions that strengthen or weaken trust in AI. Such work can help guide policy development and support more responsible innovation in higher education.

The findings emphasize that the potential of AI to enhance educational practice will be realized when students' perceptions of usefulness are supported by strong attitudes, informed awareness, and a sense of trust in the technology they use.

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